to frequency, Planck's constant therefore has units of energy times time. Planck himself realized that his new quantity could be combined with the other fundamental constants—*G*, Newton's constant of gravity, and *c*, the speed of light—to form universally defined measures of length, time, and so forth. The Planck length is about 10^{-33} centimeters, while the Planck time is about 10^{-43} seconds. The Planck length is a very short distance indeed, but presumably it has physical relevance, as a scale at which quantum mechanics (*h*), gravity (*G*), and relativity (*c*) all simultaneously matter.

Amusingly, Planck's mind immediately went to the possibility of communicating with alien civilizations. If we someday start chatting with extraterrestrial beings using interstellar radio signals, they won't know what we mean if we were to say human beings are "about two meters tall." But since they will presumably know at least as much about physics as we do, they should be aware of Planck units. This suggestion hasn't yet been put to practical use, but Planck's constant has had an immense impact elsewhere.

The idea that light is emitted in discrete quanta of energy related to its frequency is puzzling, when you think about it. From what we intuitively know about light, it might make sense if someone suggested that the amount of energy it carried depended on how *bright* it was, but not on what *color* it was. But the assumption led Planck to derive the right formula, so something about the idea seemed to be working.

It was left to Albert Einstein, in his singular way, to brush away conventional wisdom and take a dramatic leap into a new way of thinking. In 1905, Einstein suggested that light was emitted only at certain energies because it literally consisted of discrete packets, not a smooth wave. Light was particles, in other words—"photons," as they are known today. This idea, that light comes in discrete, particle-like quanta of energy, was the true birth of quantum mechanics, and was the discovery for which Einstein was awarded the Nobel Prize in 1921. (He deserved to win at least one more Nobel for the theory of relativity, but never did.) Einstein was no dummy, and he knew that this was a big deal; as he told his friend Conrad Habicht, his light quantum proposal was "very revolutionary."

Note the subtle difference between Planck's suggestion and Einstein's. Planck says that light of a fixed frequency is *emitted* in certain energy amounts, while Einstein says that's because light literally *is* discrete particles. It's the difference between saying that a certain coffee machine makes exactly one cup at a time, and saying that coffee only exists in the